

# **Milestone-1 Report: Data Preprocessing for Chest X-ray Images**

## **Project: CliniScan – Medical Image Preprocessing Pipeline**

### **Part 1: Dataset Description**

#### **1. Introduction**

Medical imaging is an essential component of modern healthcare, enabling doctors to diagnose and monitor various diseases efficiently. Among different imaging modalities, chest X-ray imaging is one of the most commonly used and cost-effective techniques for examining lung and thoracic conditions. With the increasing adoption of artificial intelligence (AI) and machine learning (ML) in healthcare, large and well-annotated datasets are required to develop reliable automated diagnostic systems. In this project, the VinBigData Chest X-ray Abnormalities Detection dataset is used as the primary data source. This dataset provides a large collection of chest X-ray images along with expert annotations, making it suitable for experimentation with data preprocessing techniques and for preparing data for future machine learning model development.

#### **2. Dataset Source**

The VinBigData Chest X-ray Abnormalities Detection dataset is publicly available on the Kaggle platform as part of a featured machine learning competition. It was released by VinBigData, a healthcare technology company, with the aim of promoting research in automated detection of abnormalities from chest X-ray images. The dataset is widely used by researchers, students, and practitioners in the fields of computer vision and medical image analysis. Its public availability and large scale make it a suitable choice for academic projects and experimental studies related to healthcare AI.

#### **3. Dataset Format and Structure**

The dataset primarily consists of chest X-ray images stored in the DICOM (Digital Imaging and Communications in Medicine) format. DICOM is a standard format used in medical imaging systems and contains not only the image data but also important metadata such as patient information, imaging parameters, and acquisition details. In addition to the image files, the dataset provides annotation files in CSV format. These annotation files include information about different types of abnormalities present in the images. For practical image processing and machine learning workflows, the DICOM images are commonly converted into standard image formats such as PNG or JPG.

#### **4. Dataset Size and Composition**

The VinBigData dataset contains approximately 15,000 chest X-ray images collected from different patients. The dataset includes both normal and abnormal cases, providing a diverse set of samples for analysis. Each image may contain one or more abnormalities, which makes the dataset suitable for multi-label classification and detection tasks. For the purpose of this project and Milestone-1 implementation, a subset of 3,000 images is used to demonstrate the complete preprocessing pipeline. This subset size is sufficient to validate the correctness, scalability, and efficiency of the preprocessing procedures.

## **5. Types of Abnormalities**

The dataset includes annotations for multiple chest abnormalities such as lung opacity, consolidation, pleural effusion, cardiomegaly, lung nodules, and other thoracic conditions. These abnormalities are clinically significant and are routinely examined by radiologists during chest X-ray interpretation. The presence of detailed annotations makes this dataset highly valuable for developing and evaluating automated systems that aim to assist medical professionals in diagnosis and decision-making.

## **6. Relevance of the Dataset to the Project**

The VinBigData Chest X-ray dataset is used in this project to implement and validate a complete data preprocessing pipeline for medical images. The dataset provides real-world, large-scale medical imaging data, which is essential for testing the robustness and practicality of preprocessing techniques before applying machine learning or deep learning models. Using this dataset ensures that the preprocessing workflow is designed to handle realistic medical image data and prepares the foundation for future stages of the project, such as feature extraction, model training, and evaluation.

## **Part 2: Data Preprocessing**

### **1. Introduction to Data Preprocessing**

Data preprocessing is a crucial step in any machine learning or image analysis pipeline. Raw data collected from real-world sources often contains inconsistencies, noise, and variations in format and size. In the case of medical images, preprocessing is especially important to ensure that the images are in a standardized and suitable form for further analysis. In this project, preprocessing is applied to chest X-ray images to convert them into a uniform format, reduce variability, and prepare them for potential use in machine learning models.

### **2. Conversion from DICOM to JPG**

The original images in the VinBigData dataset are stored in the DICOM format, which is commonly used in medical imaging systems. Although DICOM is rich in metadata, it is not always convenient for direct use in standard computer vision pipelines. Therefore, the first preprocessing step involves converting DICOM images into a standard image format such as JPG. This conversion simplifies image handling, reduces storage requirements, and allows the use of widely available image processing libraries such as OpenCV and PIL. During conversion, the pixel values are normalized and saved in JPG format while preserving the visual content of the original image.

### **3. Grayscale Conversion**

Chest X-ray images are inherently grayscale in nature, as they represent intensity variations of X-ray absorption in different tissues. Converting images to grayscale ensures that the data is represented in a single channel, which reduces computational complexity and memory usage. Using grayscale images is sufficient for most medical imaging tasks involving X-rays, as color information does not provide additional diagnostic value in this context.

### **4. Image Resizing**

Medical images in the dataset may have varying resolutions and dimensions. To ensure uniformity and compatibility with machine learning models, all images are resized to a fixed resolution of  $224 \times 224$  pixels. This size is commonly used in deep learning architectures and provides a good balance between computational efficiency and preservation of important visual details. Resizing ensures that all images have the same shape, which is a necessary requirement for batch processing and model training.

### **5. Pixel Value Normalization**

Normalization is the process of scaling pixel intensity values to a standard range, typically between 0 and 1. In this project, pixel values are divided by 255 to convert them from the

original range of 0 to 255 into a normalized range. Normalization helps improve numerical stability during model training and ensures that different images have comparable intensity distributions. It also speeds up the convergence of learning algorithms in machine learning applications.

## **6. Saving the Preprocessed Images**

After applying grayscale conversion, resizing, and normalization, the processed images are saved into a separate directory as preprocessed image files. Storing preprocessed images allows the data to be reused easily in later stages of the project without repeating the preprocessing steps. This separation between raw data and preprocessed data improves the organization of the project and supports reproducibility of experiments.

## **7. Importance of Data Preprocessing**

Data preprocessing plays a vital role in the overall performance and reliability of machine learning systems. Proper preprocessing reduces noise, removes inconsistencies, and ensures that the input data is in a suitable format for analysis. In medical imaging applications, careful preprocessing is especially important, as it directly affects the quality of the features extracted and the accuracy of the final predictions. By implementing a systematic preprocessing pipeline in this project, the foundation is established for building accurate and efficient machine learning models in future stages of development.

## **8. Conclusion**

In this Milestone-1 phase, a complete preprocessing pipeline for chest X-ray images has been implemented. The pipeline includes conversion from DICOM to JPG, grayscale conversion, resizing, normalization, and saving of preprocessed images. These steps ensure that the dataset is transformed into a clean, consistent, and machine-learning-ready format. This work provides a strong foundation for subsequent phases of the project, where advanced techniques such as feature extraction, model training, and performance evaluation can be applied to the preprocessed data.